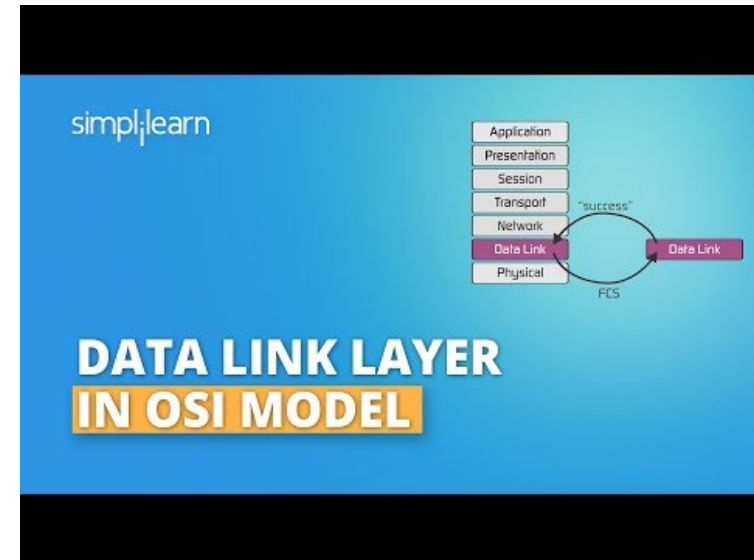


Data-Link Layer

Computer Networks

BSCS 6th
Lecture 05



Mr. Farooq Faisal

What is the Data-Link Layer?

The “Data-Link layer,” is responsible for maintaining and terminating the established connection between the network devices over the network channel.

The Data-Link Layer has two sublayers:

- The first is the “medium access control,” which uses the MAC addresses from the network devices to transmit data between them.
- The second layer is the “logical link layer,” which identifies, checks flow control, and performs the error check for the transmitted data.

Working Of the Data-Link Layer

- The data flow between the data-link layer and other layers in the OSI Model, i.e., to begin with, the network layer will share the data packets with the data-link layer.
- The data-link layer handles these data packets by integrating them with the frame structure. The frame acts as a header for the data packet, containing information about the destination address, sender address, and other related services.
- The final data format in the data-link layer is known as the data frame, which is then transmitted to the physical layer of the OSI Model.

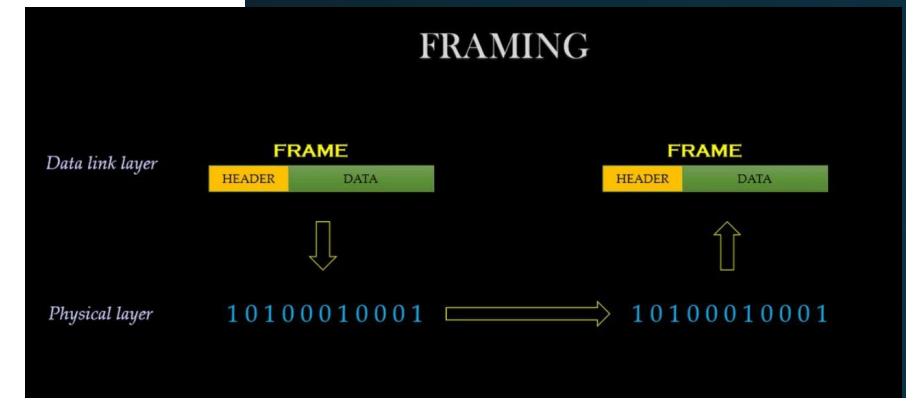
Functions of the Data-Link Layer

The working of the data-link layer in the OSI model requires the need functioning of the following attributes:

- 1) Framing
- 2) Addressing
- 3) Flow Control
- 4) Access Control
- 5) Error Control

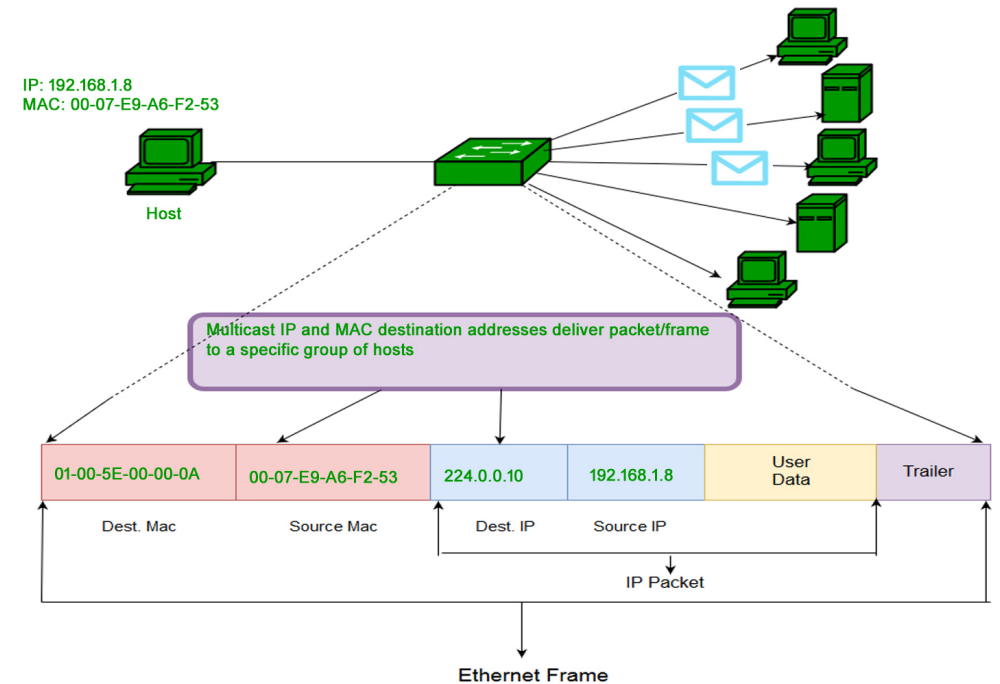
Framing

- The data packets received from the network layer are encapsulated in frames by the data-link layer for bit-to-bit sharing over the channel.
- It is also responsible for restructuring the framed data in the network model, and each data frame is different from the others.



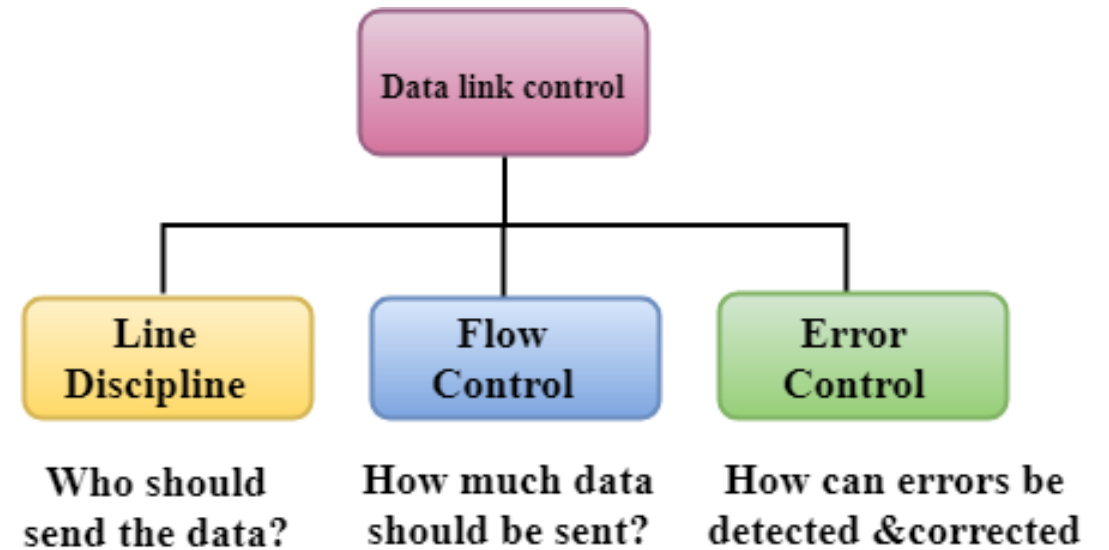
Addressing

- The task of adding a physical address to the frame in the header format is known as addressing.
- It acts as an identification service for transmitting the frames to multiple network models over the channel.



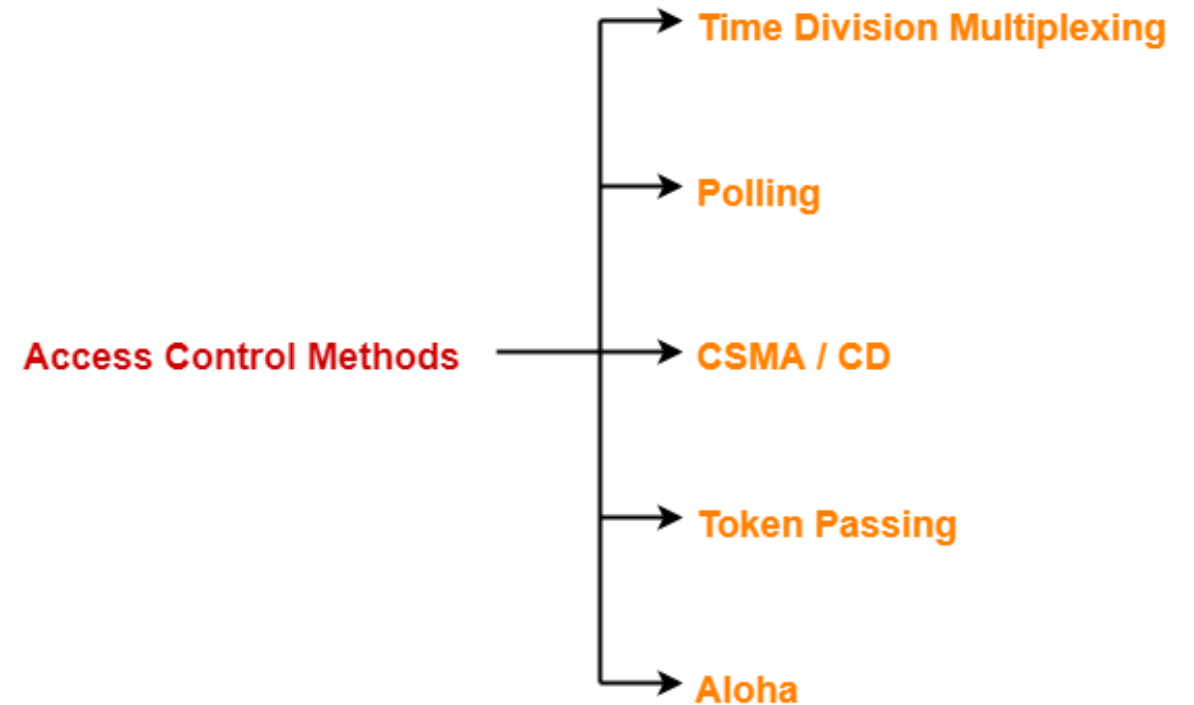
Flow Control

- During data transmission, the sender or receiver's data flow may differ, causing network traffic in the channel.
- The Data-link layer in such situations acts as a flow control for the sender side to prevent data overflow at the receiver side.



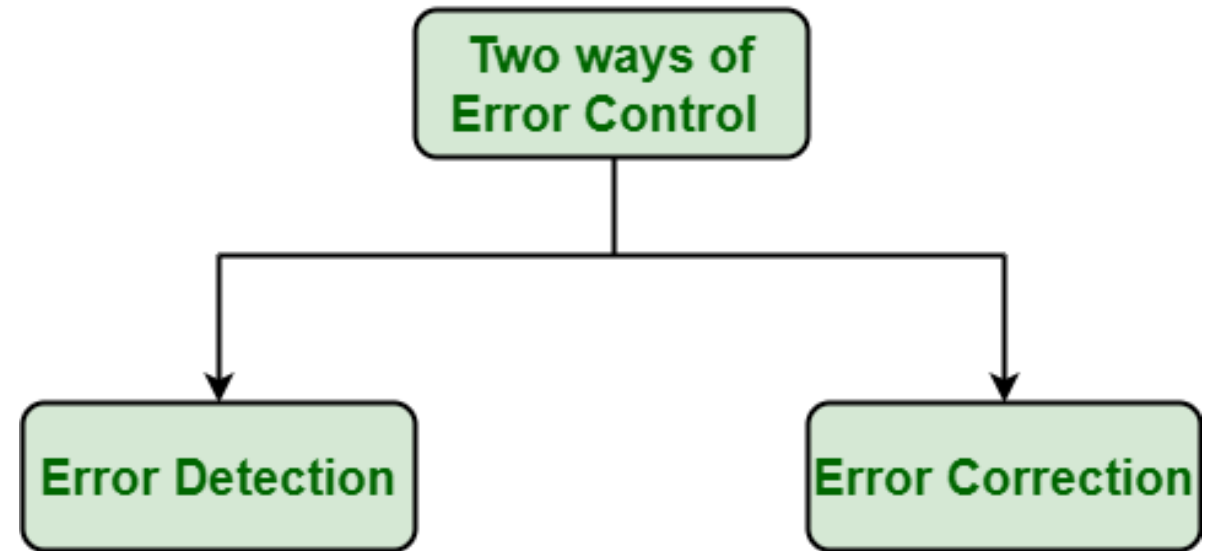
Access Control

- In this network model, when multiple devices share the same communication channel, this leads to data collision in the network channel.
- To prevent such data collision, the data-link layer performs checks on the devices with the same network channel to avoid data loss.



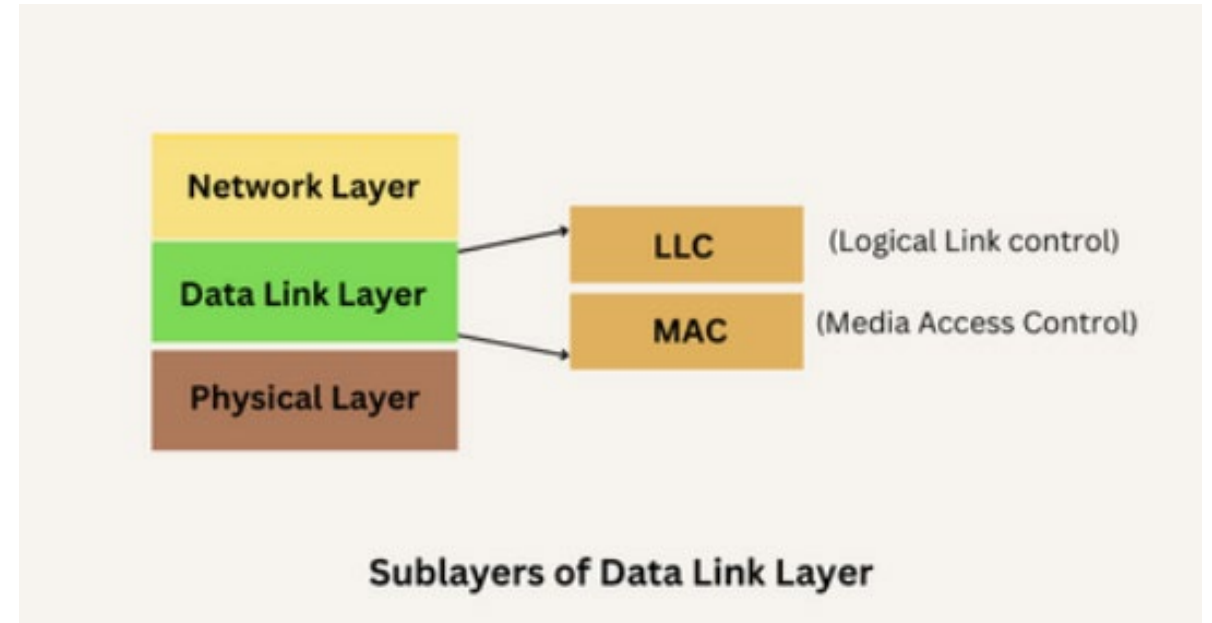
Error Control

- During data transmission, due to noise or signal loss, errors might occur in the data being transmitted.
- To minimize such data error rate, the data-link layer performs error detection and correction techniques on the transmitted data.



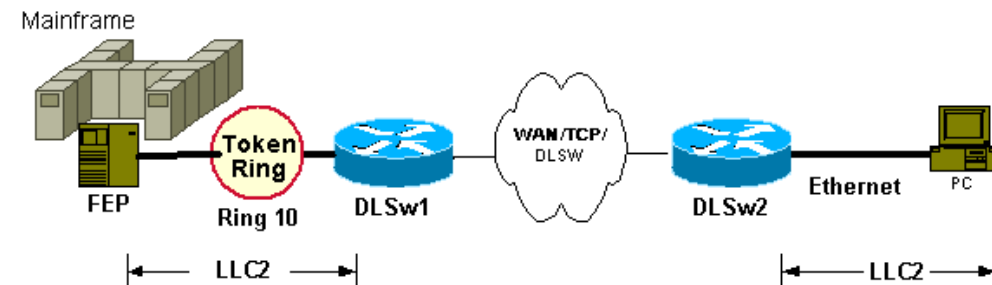
Sub-Layers of the Data-Link Layer

- The data-link layer depending on the communication with the other OSI model is divided into two types of sub-layers:
 - 1) Logical Link Control (LLC)
 - 2) Media Access Control (MAC)



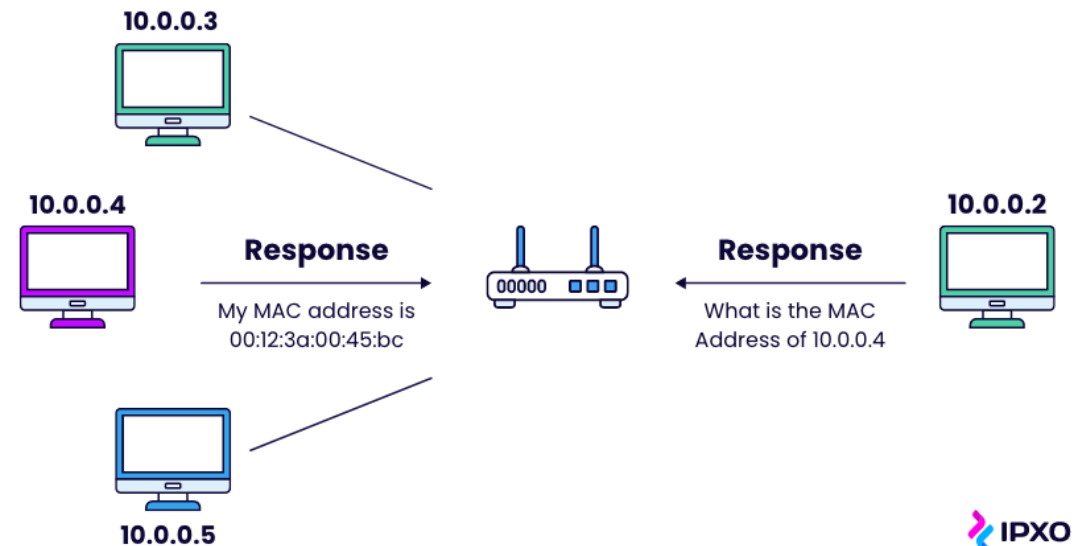
Logical Link Control (LLC)

- This is the upper sub-layer of the data-link layer.
 - The LLC sub-layer is responsible for handling and maintaining the communication between the other layers of the OSI Model.
 - It is also responsible for handling error messages and reliability checks for the data.



Media Access Control (MAC)

- This is the lower sub-layer of the data-link layer.
 - The MAC sub-layer is responsible for framing the data received from the upper layers.
 - It also is responsible for data encapsulation and media access control for the data received



Thank you

